Reconstruction Metric - GPU Optimized

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```python
# Configuración para visualización en PDF
import pandas as pd
import numpy as np
import os
import matplotlib.pyplot as plt
import subprocess

from sae.tools.experiment_utils import get_report_name
from sae.tools.experiment_utils import get_metrics_dir

pd.set_option('display.max_colwidth', None)
pd.set_option('display.max_columns', None)
pd.set_option('display.width', 100)
np.set_printoptions(linewidth=100, edgeitems=3)

os.environ['COLUMNS'] = '100'

print(" Configuración para PDF lista")
```

```
 Configuración para PDF lista
```

```python
NAMING_META = {
    'variante': 'topk',
    'tecnica': 'batch-topk',
    'capa': 6,
    'juegos': 5000,
    'base_path': 'C:\\Users\\Esposa\\Documents\\Repos\\sae-othello-gpt',
    'experiment_base_path': os.path.abspath('../../experiments'),
    'extras': {
        'expansion': 8,
        'k': 50,
        'epochs': 1000,
        'patience': 20
    }
}
print('\n Metadata de naming:')
```

```
for key, value in NAMING_META.items():
    print(f'  {key}: {value}')
```

```
 Metadata de naming:
  variante: topk
  tecnica: batch-topk
  capa: 6
  juegos: 5000
  base_path: C:\Users\Esposa\Documents\Repos\sae-othello-gpt
  experiment_base_path: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments
  extras: {'expansion': 8, 'k': 50, 'epochs': 1000, 'patience': 20}
```

0.1 1. Setup e Imports

```
import numpy as np
import torch
import torch.nn as nn
import torch.nn.functional as F
import sys
from pathlib import Path
from tqdm import tqdm
import time

from sae.tools.naming_utils import get_checkpoint_dir, get_model_name, get_extras_id

project_root = Path('../../..').resolve()
sys.path.insert(0, str(project_root))

device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(f"Device: {device}")
if torch.cuda.is_available():
    print(f"GPU: {torch.cuda.get_device_name(0)}")
    print(f"Memoria: {torch.cuda.get_device_properties(0).total_memory / 1e9:.2f} GB")
```

```
Device: cuda
GPU: NVIDIA GeForce RTX 5060
Memoria: 8.55 GB
```

```
class BatchTopKSAE(nn.Module):
    def __init__(self, d_in: int, d_sae: int, k: int, k_aux: int = 512):
        super().__init__()
        self.d_in  = d_in
        self.d_sae = d_sae
        self.k     = k
        self.k_aux = k_aux

        self.W_enc = nn.Parameter(torch.empty(d_in, d_sae))
        self.W_dec = nn.Parameter(torch.empty(d_sae, d_in))
        self.b_dec = nn.Parameter(torch.zeros(d_in))

        nn.init.kaiming_uniform_(self.W_enc)
        nn.init.kaiming_uniform_(self.W_dec)
        self.W_dec.data = self.W_dec.data / self.W_dec.data.norm(dim=-1, keepdim=True)
        self.register_buffer('num_batches_not_active', torch.zeros(d_sae))

    def encode(self, x: torch.Tensor, threshold=None) -> torch.Tensor:
        x_norm, _, _ = self.preprocess_input(x)
        x_cent   = x_norm - self.b_dec
```

```
        pre_acts = F.relu(x_cent @ self.W_enc)
        return self._topk_activation(pre_acts, threshold=threshold)

    def _topk_activation(self, x: torch.Tensor, threshold=None) -> torch.Tensor:
        if threshold is None:
            acts_topk = torch.topk(x.flatten(), self.k * x.shape[0], dim=-1)
            return (
                torch.zeros_like(x.flatten())
                .scatter(-1, acts_topk.indices, acts_topk.values)
                .reshape(x.shape)
            )
        return torch.where(x > threshold, x, torch.zeros_like(x))

    def decode(self, hidden: torch.Tensor) -> torch.Tensor:
        return hidden @ self.W_dec + self.b_dec

    def preprocess_input(self, x: torch.Tensor):
        x_mean = x.mean(dim=-1, keepdim=True)
        x = x - x_mean
        x_std = x.std(dim=-1, keepdim=True)
        x = x / (x_std + 1e-5)
        return x, x_mean, x_std

    def postprocess_output(self, x_reconstruct, x_mean, x_std):
        return x_reconstruct * x_std + x_mean

    def forward(self, x: torch.Tensor, threshold=None):
        x_norm, x_mean, x_std = self.preprocess_input(x)
        x_cent   = x_norm - self.b_dec
        pre_acts = F.relu(x_cent @ self.W_enc)
        hidden   = self._topk_activation(pre_acts, threshold=threshold)
        recon    = self.postprocess_output(self.decode(hidden), x_mean, x_std)
        return recon, hidden, pre_acts

print("BatchTopKSAE definida")
```

```
BatchTopKSAE definida
```

0.2 2. Cargar Datos

```
# Cargar activaciones
activations_path = project_root / "sae" / "activations" / "data" / "layer6_2000games.npy"
activations = np.load(activations_path)

print(f"Activaciones del modelo:")
print(f"  Shape: {activations.shape}")
print(f"  Memoria: {activations.nbytes / (1024**2):.2f} MB")
```

```
Activaciones del modelo:
  Shape: (118000, 512)
  Memoria: 230.47 MB
```

```
# Cargar ground truth de BSPs
bsp_gt_path = project_root / "sae" / "metrics" / "02_data" / "bsp_ground_truth_2000games.npy"
bsp_names_path = project_root / "sae" / "metrics" / "02_data" / "bsp_ground_truth_2000games.names.npy"

bsp_ground_truth = np.load(bsp_gt_path)
bsp_names = np.load(bsp_names_path, allow_pickle=True)
```

```python
print(f"\nGround truth BSPs:")
print(f"  Shape: {bsp_ground_truth.shape}")
print(f"  Total BSPs: {len(bsp_names)}")
```

```
Ground truth BSPs:
  Shape: (118000, 326)
  Total BSPs: 326
```

0.3 3. Cargar SAE y Extraer Features

```python
input_dim = 512
hidden_dim = 4096

sae = BatchTopKSAE(d_in=input_dim, d_sae=hidden_dim, k=NAMING_META['extras']['k']).to(device)
checkpoint_dir = get_checkpoint_dir(
    NAMING_META['base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos']
)
extras_id = get_extras_id(checkpoint_dir, NAMING_META['extras']) if NAMING_META.get('extras') else None

model_filename = get_model_name(
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    'best',
    extras_id=extras_id
)
model_path = checkpoint_dir / model_filename
checkpoint = torch.load(model_path, map_location=device, weights_only=False)
sae.load_state_dict(checkpoint['model_state_dict'])
sae.eval()

history = checkpoint.get('history', {})
best_val_mse = history.get('val_mse_loss', [float('nan')])
best_val_mse = min(best_val_mse) if best_val_mse else float('nan')

print(f"SAE cargado:")
print(f"  Path: {model_path}")
print(f"  Input: {input_dim}, Hidden: {hidden_dim}, k: {NAMING_META['extras']['k']}")
print(f"  Expansion: {hidden_dim/input_dim}x")
print(f"  Best Val MSE: {best_val_mse:.6f}")
print(f"  Extras ID: {extras_id}")
```

```
SAE cargado:
  Path: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk\sae_batch-topk_topk\5000games\sae_topk
topk_l6_5000g_ex1_best.pt
  Input: 512, Hidden: 4096, k: 50
  Expansion: 8.0x
  Best Val MSE: 0.174040
  Extras ID: ex1
```

```python
# Extraer features del SAE en GPU
print("Extrayendo features del SAE...")
activations_tensor = torch.from_numpy(activations).float().to(device)
```

```
with torch.no_grad():
    sae_features = sae.encode(activations_tensor)

print(f"\nFeatures SAE:")
print(f"  Shape: {sae_features.shape}")
print(f"  Device: {sae_features.device}")
print(f"  Sparsity: {(sae_features == 0).float().mean():.2%}")
print(f"  Activaciones promedio: {(sae_features > 0).sum(dim=1).float().mean():.1f}")
```

```
Extrayendo features del SAE...

Features SAE:
  Shape: torch.Size([118000, 4096])
  Device: cuda:0
  Sparsity: 98.78%
  Activaciones promedio: 50.0
```

0.4 4. Split Train/Test y Filtrar BSPs de Piezas

```
# Split 50/50:
n_moves = 59
n_games = 2000  # número de partidas válidas
train_games = n_games // 2
split_idx = train_games * n_moves

sae_features_train = sae_features[:split_idx]
sae_features_test  = sae_features[split_idx:]

print(f"Total partidas: {n_games}")
print(f"Train: {train_games} partidas ({split_idx} activaciones)")
print(f"Test:  {n_games - train_games} partidas ({len(activations) - split_idx} activaciones)")
print(f"Train set: {sae_features_train.shape}")
print(f"Test set:  {sae_features_test.shape}")
```

```
Total partidas: 2000
Train: 1000 partidas (59000 activaciones)
Test:  1000 partidas (59000 activaciones)
Train set: torch.Size([59000, 4096])
Test set:  torch.Size([59000, 4096])
```

```
# ============================================
# CONFIGURACIÓN DE FILTRADO DE BSPs
# ============================================
USE_PLAYER   = True   # BSPs perspectiva del jugador (1 y 2)
USE_ABSOLUTE = False    # BSPs absolutas negro/blanco (B y W)
USE_STRATEGIC = False  # BSPs especiales (BSP_)

bsp_pieces_indices = []
bsp_pieces_names = []

for i, name in enumerate(bsp_names):
    include = False
    if USE_PLAYER and len(name) == 6 and name.startswith('BSP') and (name.endswith('1') or name.endswith('2')):
        include = True
    if USE_ABSOLUTE and (name.endswith('B') or name.endswith('W')):
        include = True
    if USE_STRATEGIC and name.startswith('BSP_'):
        include = True
    if include:
```

```
        bsp_pieces_indices.append(i)
        bsp_pieces_names.append(name)

bsp_pieces_indices = np.array(bsp_pieces_indices)

# Para reconstruction necesitas split train/test
bsp_gt_train_pieces = torch.from_numpy(bsp_ground_truth[:split_idx, bsp_pieces_indices]).bool().to(device)
bsp_gt_test_pieces  = torch.from_numpy(bsp_ground_truth[split_idx:, bsp_pieces_indices]).bool().to(device)

parts = []
if USE_PLAYER: parts.append("player")
if USE_ABSOLUTE: parts.append("absolute")
if USE_STRATEGIC: parts.append("strategic")
filter_suffix = "_".join(parts) if parts else "none"

print(f"BSPs seleccionadas para Reconstruction:")
print(f"  Player: {USE_PLAYER} | Absolute: {USE_ABSOLUTE} | Strategic: {USE_STRATEGIC}")
print(f"  Total: {len(bsp_pieces_indices)}")
print(f"  Train shape: {bsp_gt_train_pieces.shape}")
print(f"  Test shape:  {bsp_gt_test_pieces.shape}")
print(f"  Device: {bsp_gt_train_pieces.device}")
print(f"  Primeras 5: {', '.join(bsp_pieces_names[:5])}")
print(f"  Últimas 5: {', '.join(bsp_pieces_names[-5:])}")
print(f"  Filter suffix: {filter_suffix}")
```

```
BSPs seleccionadas para Reconstruction:
  Player: True | Absolute: False | Strategic: False
  Total: 128
  Train shape: torch.Size([59000, 128])
  Test shape:  torch.Size([59000, 128])
  Device: cuda:0
  Primeras 5: BSPA11, BSPA12, BSPA21, BSPA22, BSPA31
  Últimas 5: BSPH62, BSPH71, BSPH72, BSPH81, BSPH82
  Filter suffix: player
```

0.5 5. Funciones GPU-Optimizadas

```
def fast_precision_score_gpu(y_true, y_pred, eps=1e-8):
    """
    Precisión vectorizada en GPU.

    Args:
        y_true: Tensor (n_positions,) booleano
        y_pred: Tensor (n_positions, n_candidates) booleano

    Returns:
        precision_scores: Tensor (n_candidates,)
    """
    y_true_expanded = y_true.unsqueeze(1)

    tp = (y_true_expanded & y_pred).sum(dim=0).float()
    fp = (~y_true_expanded & y_pred).sum(dim=0).float()

    precision = tp / (tp + fp + eps)

    return precision


def fast_f1_score_gpu(y_true, y_pred, eps=1e-8):
    """
```

```
    F1 score vectorizado en GPU.

    Args:
        y_true: Tensor (n_positions,) booleano
        y_pred: Tensor (n_positions,) booleano

    Returns:
        f1: Float
    """
    tp = (y_true & y_pred).sum().float()
    fp = (~y_true & y_pred).sum().float()
    fn = (y_true & ~y_pred).sum().float()

    precision = tp / (tp + fp + eps)
    recall = tp / (tp + fn + eps)

    f1 = 2 * (precision * recall) / (precision + recall + eps)

    return f1.item()

print(" Funciones GPU definidas")
```

```
 Funciones GPU definidas
```

0.6 6. Fase 1: Identificar Features de Alta Precisión (GPU)

Para cada BSP, encontrar features con precisión   0.95 en el train set.

```
def identify_high_precision_features_gpu(
    sae_features_train,
    bsp_gt_train,
    f_max,
    precision_threshold=0.95,
    significance_threshold=1,
    thresholds=[0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9],
    feature_batch_size=512,
    verbose=True
):
    """
    Para cada (threshold t, feature f, BSP b): determina si f es clasificador
    de alta precisión para b al threshold t.

    Criterios (igual que implementación oficial):
        - precision(f, b, t) >= precision_threshold   (default 0.95)
        - on_count(f, t)     >= significance_threshold (default 10)

    Args:
        sae_features_train:    Tensor (P, n_features) GPU
        bsp_gt_train:          Tensor (P, n_bsps) GPU, bool
        f_max:                 Tensor (n_features,) GPU - calculado FUERA para
                               garantizar identidad exacta con Fase 2
        precision_threshold:   float, default 0.95
        significance_threshold: int, mín. activaciones requeridas (default 10)
        thresholds:            lista de fracciones t   [0, 1)
        feature_batch_size:    int, features por mini-batch interno

    Returns:
        hp_mask_TFB:            BoolTensor (T, n_active, n_bsps)
        active_feature_indices: LongTensor (n_active,)
    """
    n_positions, n_features = sae_features_train.shape
```

```python
    n_bsps = bsp_gt_train.shape[1]
    device = sae_features_train.device

    thresholds_T = torch.tensor(thresholds, device=device)
    n_thresholds = len(thresholds_T)

    # Features vivas (f_max > 0)
    active_feature_indices = (f_max > 0).nonzero(as_tuple=True)[0]
    n_active = len(active_feature_indices)

    if verbose:
        print("=" * 60)
        print("FASE 1: IDENTIFICAR FEATURES DE ALTA PRECISIÓN")
        print("=" * 60)
        print(f"Precision threshold:     {precision_threshold}")
        print(f"Significance threshold:  >= {significance_threshold} activaciones")
        print(f"Features activas:        {n_active}/{n_features} ({n_active/n_features*100:.1f}%)")
        print(f"BSPs:                    {n_bsps}")
        print(f"Thresholds:              {thresholds}")
        print("=" * 60)

    # Thresholds absolutos: thresh_TF[t, f] = t * f_max[f]
    active_f_max  = f_max[active_feature_indices]           # (n_active,)
    thresholds_TF = thresholds_T[:, None] * active_f_max   # (T, n_active)

    # Acumuladores sobre todo el train set
    on_count_TF  = torch.zeros(n_thresholds, n_active, device=device)
    tp_count_TFB = torch.zeros(n_thresholds, n_active, n_bsps, device=device)

    bsp_float_PB = bsp_gt_train.float()  # (P, B)

    n_iters = (n_active + feature_batch_size - 1) // feature_batch_size

    for fi in tqdm(range(n_iters), desc="Fase 1"):
        f_start = fi * feature_batch_size
        f_end   = min(f_start + feature_batch_size, n_active)

        global_idx = active_feature_indices[f_start:f_end]
        acts_PF    = sae_features_train[:, global_idx]     # (P, bs)
        thresh_TF  = thresholds_TF[:, f_start:f_end]      # (T, bs)

        # active_TFP[t, f, p] = True si acts[p,f] > thresh[t,f]
        active_TFP = acts_PF.T.unsqueeze(0) > thresh_TF.unsqueeze(2)  # (T, bs, P)

        # Conteo de activaciones por (threshold, feature)
        on_count_TF[:, f_start:f_end] += active_TFP.sum(dim=2).float()

        # True positives: f activa Y BSP verdadera
        # tp[t, f, b] = Σ_p active[t,f,p] * bsp[p,b]
        tp_count_TFB[:, f_start:f_end, :] += torch.einsum(
            'tfp,pb->tfb', active_TFP.float(), bsp_float_PB
        )

    # Precisión: tp / on_count
    eps = 1e-8
    precision_TFB = tp_count_TFB / (on_count_TF.unsqueeze(2) + eps)  # (T, F, B)

    # Máscara: alta precisión AND significancia suficiente
    sig_mask_TF1 = (on_count_TF >= significance_threshold).unsqueeze(2)  # (T, F, 1)
    hp_mask_TFB  = (precision_TFB >= precision_threshold) & sig_mask_TF1  # (T, F, B)

    if verbose:
```

```python
        hp_per_t = hp_mask_TFB.sum(dim=(1, 2))  # (T,)
        hp_per_b = hp_mask_TFB.sum(dim=1)        # (T, B)
        best_t   = hp_per_t.argmax().item()
        print(f"\nPares (feature, BSP) de alta precisión por threshold:")
        for i, t in enumerate(thresholds):
            marker = " ← más pares" if i == best_t else ""
            print(f"  t={t:.1f}: {hp_per_t[i].item():5d} pares{marker}")
        print(f"\nFeatures HP por BSP en t={thresholds[best_t]:.1f}:")
        print(f"  Media:             {hp_per_b[best_t].float().mean():.1f}")
        print(f"  Máx:               {hp_per_b[best_t].max().item()}")
        print(f"  BSPs sin features: {(hp_per_b[best_t] == 0).sum().item()}")

    return hp_mask_TFB, active_feature_indices

print("  Fase 1 rediseñada: umbral de significancia ( 10) + estructura (T, F, B)")
```

```
 Fase 1 rediseñada: umbral de significancia ( 10) + estructura (T, F, B)
```

```python
THRESHOLDS = [0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9]

# f_max calculado UNA SOLA VEZ aquí - se pasa a Fase 1 y Fase 2
f_max = sae_features_train.max(dim=0)[0]  # (n_features,)
print(f"f_max calculado: {(f_max > 0).sum().item()} features activas de {f_max.shape[0]}")

start_time = time.time()

hp_mask_TFB, active_feature_indices = identify_high_precision_features_gpu(
    sae_features_train,
    bsp_gt_train_pieces,
    f_max,
    precision_threshold=0.95,
    significance_threshold=1,
    thresholds=THRESHOLDS,
    feature_batch_size=512,
    verbose=True
)

phase1_time = time.time() - start_time

print(f"\nTiempo Fase 1: {phase1_time:.2f} seg")
print(f"hp_mask_TFB shape: {hp_mask_TFB.shape}  (T, n_active, n_bsps)")
```

```
f_max calculado: 4096 features activas de 4096
============================================================
FASE 1: IDENTIFICAR FEATURES DE ALTA PRECISIÓN
============================================================
Precision threshold:     0.95
Significance threshold:  >= 1 activaciones
Features activas:        4096/4096 (100.0%)
BSPs:                    128
Thresholds:              [0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9]
============================================================

Fase 1: 100%|     | 8/8 [00:01<00:00,  6.48it/s]


Pares (feature, BSP) de alta precisión por threshold:
  t=0.0:    61 pares
  t=0.1:    70 pares
  t=0.2:   638 pares
  t=0.3:  1573 pares
```

```
  t=0.4:  2508 pares
  t=0.5:  3502 pares
  t=0.6:  4890 pares
  t=0.7:  8492 pares
  t=0.8: 19626 pares
  t=0.9: 51030 pares ← más pares

Features HP por BSP en t=0.9:
  Media:             398.7
  Máx:               1420
  BSPs sin features: 0

Tiempo Fase 1: 10.08 seg
hp_mask_TFB shape: torch.Size([10, 4096, 128])  (T, n_active, n_bsps)
```

0.7 7. Fase 2: Reconstruir Tableros en Test Set (GPU)

```python
def reconstruct_boards_gpu(
    sae_features_test,
    bsp_gt_test,
    hp_mask_TFB,
    active_feature_indices,
    f_max,
    thresholds=[0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9],
    verbose=True
):
    """
    Fase 2: barrido simultáneo de thresholds T → toma el mejor F1.

    Para cada threshold t:
      1. Binarizar features de test: active[p, f] = acts[p, f] > t * f_max[f]
      2. Predicción por (posición, BSP):
            predict[p, b] = OR_f ( active[p,f] AND hp_mask[t, f, b] )
            implementado como: (active_PF @ hp_FB) > 0
      3. F1 global micro-averaged (acumula TP/FP/FN sobre todas las posiciones y BSPs)

    Retorna el max F1 sobre todos los thresholds (Eq. 5 del paper: max_t).

    Args:
        sae_features_test:      Tensor (P, n_features) GPU
        bsp_gt_test:            Tensor (P, n_bsps) GPU, bool
        hp_mask_TFB:            BoolTensor (T, n_active, n_bsps) de Fase 1
        active_feature_indices: LongTensor (n_active,) de Fase 1
        f_max:                  Tensor (n_features,) GPU - MISMO objeto que Fase 1
        thresholds:             lista de fracciones (debe coincidir con Fase 1)

    Returns:
        best_f1:          float, max F1 sobre thresholds
        best_threshold:   float, threshold óptimo
        f1_per_threshold: np.array (T,), F1 por threshold
    """
    device = sae_features_test.device
    n_thresholds = len(thresholds)
    thresholds_T = torch.tensor(thresholds, device=device)

    # Activaciones del test solo para features activas
    test_active_PF = sae_features_test[:, active_feature_indices]  # (P, n_active)
    active_f_max   = f_max[active_feature_indices]                  # (n_active,)

    gt_PB = bsp_gt_test.bool()  # (P, B)
    eps   = 1e-8
```

```
    f1_scores   = torch.zeros(n_thresholds, device=device)
    precision_T = torch.zeros(n_thresholds, device=device)
    recall_T    = torch.zeros(n_thresholds, device=device)

    if verbose:
        print("=" * 60)
        print("FASE 2: RECONSTRUCCIÓN - BARRIDO DE THRESHOLDS")
        print("=" * 60)
        print(f"{'t':>5}  {'F1':>8}  {'P':>8}  {'R':>8}  {'TP':>8}  {'FP':>8}  {'FN':>8}")
        print("-" * 60)

    for t_idx in range(n_thresholds):
        t = thresholds[t_idx]

        # Threshold absoluto por feature: t * f_max[f]
        abs_thresh_F = thresholds_T[t_idx] * active_f_max  # (n_active,)

        # Binarizar features en test
        active_PF = test_active_PF > abs_thresh_F.unsqueeze(0)  # (P, n_active)

        # predict[p, b] = any( active[p,f] AND hp[t,f,b] )
        # = (active_PF.float() @ hp_mask[t].float()) > 0
        hp_FB          = hp_mask_TFB[t_idx].float()              # (n_active, B)
        predictions_PB = (active_PF.float() @ hp_FB) > 0         # (P, B)

        # Métricas globales micro-averaged
        tp = (predictions_PB &  gt_PB).sum().float()
        fp = (predictions_PB & ~gt_PB).sum().float()
        fn = (~predictions_PB & gt_PB).sum().float()

        p  = tp / (tp + fp + eps)
        r  = tp / (tp + fn + eps)
        f1 = 2 * p * r / (p + r + eps)

        f1_scores[t_idx]   = f1
        precision_T[t_idx] = p
        recall_T[t_idx]    = r

        if verbose:
            print(f"{t:>5.1f}  {f1.item():>8.4f}  {p.item():>8.4f}  {r.item():>8.4f}"
                  f"  {tp.item():>8.0f}  {fp.item():>8.0f}  {fn.item():>8.0f}")

    best_idx       = f1_scores.argmax().item()
    best_f1        = f1_scores[best_idx].item()
    best_threshold = thresholds[best_idx]

    if verbose:
        print("=" * 60)
        print(f"Mejor threshold: t={best_threshold:.1f}")
        print(f"Reconstruction Score (max_t F1): {best_f1:.4f}")
        print("=" * 60)

    return best_f1, best_threshold, f1_scores.cpu().numpy()

print("  Fase 2 rediseñada: barrido T simultáneo + F1 global micro-averaged + max_t")
```

```
  Fase 2 rediseñada: barrido T simultáneo + F1 global micro-averaged + max_t
```

```
start_time = time.time()

reconstruction_score, best_threshold, f1_per_threshold = reconstruct_boards_gpu(
```

```
    sae_features_test,
    bsp_gt_test_pieces,
    hp_mask_TFB,
    active_feature_indices,
    f_max,           # mismo tensor calculado antes de Fase 1
    thresholds=THRESHOLDS,
    verbose=True
)

phase2_time = time.time() - start_time
total_time  = phase1_time + phase2_time

print(f"\nTiempo Fase 2: {phase2_time:.2f} seg")
print(f"Tiempo total:  {total_time:.2f} seg ({total_time/60:.2f} min)")
```

```
============================================================
FASE 2: RECONSTRUCCIÓN - BARRIDO DE THRESHOLDS
============================================================
    t        F1         P         R        TP        FP        FN
------------------------------------------------------------
  0.0    0.0078    0.9293    0.0039      7859       598   1998141
  0.1    0.0084    0.9289    0.0042      8486       650   1997514
  0.2    0.0768    0.9699    0.0400     80247      2493   1925753
  0.3    0.1791    0.9697    0.0987    197943      6175   1808057
  0.4    0.2030    0.9799    0.1133    227184      4652   1778816
  0.5    0.1593    0.9826    0.0867    173856      3085   1832144
  0.6    0.1091    0.9747    0.0578    115943      3014   1890057
  0.7    0.0745    0.9240    0.0388     77816      6399   1928184
  0.8    0.0584    0.7644    0.0304     60892     18770   1945108
  0.9    0.0623    0.5949    0.0329     65984     44930   1940016
============================================================
Mejor threshold: t=0.4
Reconstruction Score (max_t F1): 0.2030
============================================================

Tiempo Fase 2: 0.30 seg
Tiempo total:  10.38 seg (0.17 min)
```

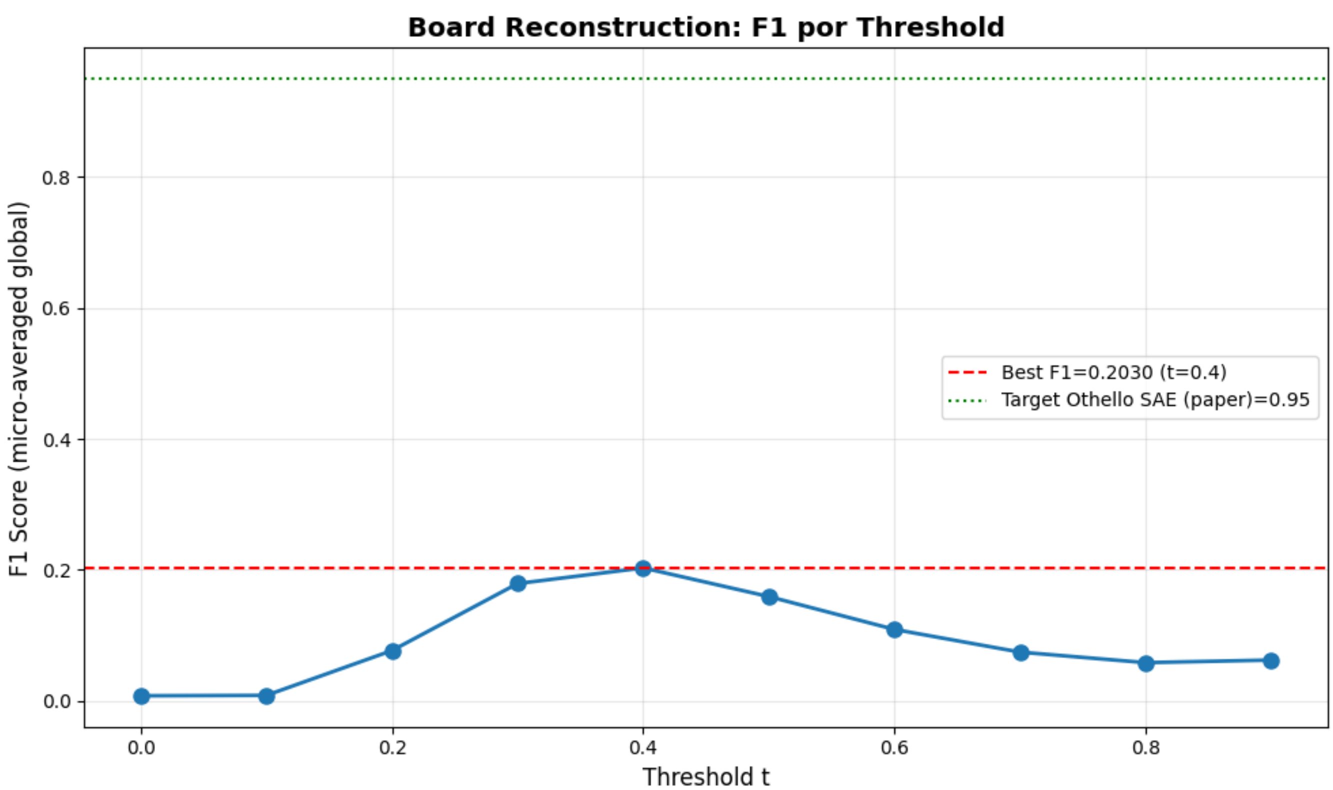
0.8 8. Análisis de Resultados

```
print("=" * 60)
print("BARRIDO DE THRESHOLDS - F1 POR THRESHOLD")
print("=" * 60)
for i, (t, f1) in enumerate(zip(THRESHOLDS, f1_per_threshold)):
    marker = " ← MEJOR" if i == f1_per_threshold.argmax() else ""
    print(f"t={t:.1f}: F1={f1:.4f}{marker}")
print("=" * 60)

plt.figure(figsize=(10, 6))
plt.plot(THRESHOLDS, f1_per_threshold, marker='o', linewidth=2, markersize=8)
plt.axhline(y=reconstruction_score, color='r', linestyle='--',
            label=f'Best F1={reconstruction_score:.4f} (t={best_threshold:.1f})')
plt.axhline(y=0.95, color='g', linestyle=':', label='Target Othello SAE (paper)=0.95')
plt.xlabel('Threshold t', fontsize=12)
plt.ylabel('F1 Score (micro-averaged global)', fontsize=12)
plt.title('Board Reconstruction: F1 por Threshold', fontsize=14, fontweight='bold')
plt.grid(True, alpha=0.3)
plt.legend()
plt.tight_layout()
plt.show()
```

```
print(f"\n  Threshold óptimo: t={best_threshold:.1f}")
print(f"  Reconstruction Score: {reconstruction_score:.4f}")
```

```
============================================================
BARRIDO DE THRESHOLDS - F1 POR THRESHOLD
============================================================
t=0.0: F1=0.0078
t=0.1: F1=0.0084
t=0.2: F1=0.0768
t=0.3: F1=0.1791
t=0.4: F1=0.2030 ← MEJOR
t=0.5: F1=0.1593
t=0.6: F1=0.1091
t=0.7: F1=0.0745
t=0.8: F1=0.0584
t=0.9: F1=0.0623
============================================================
```




```
  Threshold óptimo: t=0.4
  Reconstruction Score: 0.2030
```

```
# Estadísticas sobre features de alta precisión por BSP
hp_per_bsp = hp_mask_TFB.sum(dim=1)  # (T, n_bsps)
best_t_idx = torch.tensor(f1_per_threshold).argmax().item()
n_features_per_bsp = hp_per_bsp[best_t_idx].cpu().numpy()

print("="*60)
print("ESTADÍSTICAS - FEATURES POR BSP")
print("="*60)
print(f"Total BSPs: {len(n_features_per_bsp)}")
print(f"Features promedio por BSP: {np.mean(n_features_per_bsp):.1f}")
```

```
print(f"Features mediana por BSP: {np.median(n_features_per_bsp):.1f}")
print(f"BSPs con 0 features: {np.sum(n_features_per_bsp == 0)}")
print(f"BSPs con 1+ features: {np.sum(n_features_per_bsp > 0)}")
print(f"BSPs con 5+ features: {np.sum(n_features_per_bsp >= 5)}")
print(f"BSPs con 10+ features: {np.sum(n_features_per_bsp >= 10)}")
print(f"Max features en una BSP: {np.max(n_features_per_bsp)}")
print("="*60)
```

```
============================================================
ESTADÍSTICAS - FEATURES POR BSP
============================================================
Total BSPs: 128
Features promedio por BSP: 19.6
Features mediana por BSP: 6.0
BSPs con 0 features: 3
BSPs con 1+ features: 125
BSPs con 5+ features: 79
BSPs con 10+ features: 46
Max features en una BSP: 176
============================================================
```

0.9 9. Guardar Resultados

```
results = {
    'reconstruction_score': reconstruction_score,
    'best_threshold': best_threshold,
    'f1_per_threshold': f1_per_threshold,
    'thresholds': np.array(THRESHOLDS),
    'phase1_time_seconds': phase1_time,
    'phase2_time_seconds': phase2_time,
    'total_time_seconds': total_time,
    'bsp_names': bsp_pieces_names,
}

metrics_dir = get_metrics_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)
output_path = metrics_dir / f"reconstruction_results_{filter_suffix}.npz"
np.savez(output_path, **results)

print(f"  Resultados guardados en: {output_path}")
print(f"\nContenido:")
print(f"  reconstruction_score: {reconstruction_score:.4f}")
print(f"  best_threshold:       {best_threshold:.1f}")
print(f"  f1_per_threshold:     {f1_per_threshold}")

# Comparación con el paper
print()
print("=" * 60)
print("COMPARACIÓN CON PAPER (Karvonen et al., NeurIPS 2024)")
print("=" * 60)
print(f"{'Modelo':<25} {'Reconstruction':>14}")
print("-" * 40)
print(f"{'SAE trained (paper)':<25} {'0.95':>14}  ← Objetivo Othello")
print(f"{'Nuestro SAE':<25} {reconstruction_score:>14.4f}  ← Resultado")
```

```python
print("=" * 60)

if reconstruction_score >= 0.90:
    print("\n  Excelente: Muy cercano al objetivo del paper")
elif reconstruction_score >= 0.50:
    print("\n~ Moderado: Por encima de random, revisar SAE")
else:
    print("\n  Bajo: Revisar entrenamiento del SAE o calidad de las activaciones")
```

```
  Resultados guardados en: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_topk\sae_batch-
topk_topk\5000games\ex1\metrics\reconstruction_results_player.npz

Contenido:
  reconstruction_score: 0.2030
  best_threshold:       0.4
  f1_per_threshold:     [0.0078026  0.00842226 0.07683771 0.17912437 0.203039   0.15928602 0.10912503 0.07445741 0.0583
 0.06233981]

============================================================
COMPARACIÓN CON PAPER (Karvonen et al., NeurIPS 2024)
============================================================
Modelo                    Reconstruction
----------------------------------------
SAE trained (paper)                 0.95  ← Objetivo Othello
Nuestro SAE                       0.2030  ← Resultado
============================================================

  Bajo: Revisar entrenamiento del SAE o calidad de las activaciones
```

0.10 10. Generar PDF con Quarto

```python
output_dir = get_metrics_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)

pdf_name = get_report_name(
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    f'reconstruction_{filter_suffix}',
    extras_id=extras_id
)

notebook_name = 'reconstruction_sae_batchtopk.ipynb'
output_path = output_dir / pdf_name

print(f' Generando PDF con Quarto...')
print(f' Archivo de salida: {output_path}')

result = subprocess.run(
    f'quarto render {notebook_name} --to pdf --output-dir "{output_dir}" --output {pdf_name}',
    shell=True,
    capture_output=True,
    text=True
```

```python
)

if result.returncode == 0:
    print(f' PDF generado exitosamente: {output_path}')
else:
    print(f' Error al generar PDF:')
    print(result.stderr)
```

```
 Generando PDF con Quarto...
 Archivo de salida: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_topk\sae_batch-
topk_topk\5000games\ex1\metrics\sae_topk_batch-topk_l6_5000g_ex1_reconstruction_player.pdf
 PDF generado exitosamente: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_topk\sae_batch-
topk_topk\5000games\ex1\metrics\sae_topk_batch-topk_l6_5000g_ex1_reconstruction_player.pdf
```